Sorting with a Weighing balance

Grade 7: Finding the Unknown

Introduction

We have used computers or smartphones. We mostly need to modify our data by Alphabetic order, Numerical order, or date modified order in ascending or descending order. And computers do it very quickly. Have you ever thought about how computers do this process very quickly? In this activity, we can use our understanding of a weighing scale to sort containers by weight. This can be done in many ways, and some of these methods will require less number of weighings than others. This is similar to how a computer also sorts.

Material Required

Weighting Balance
7 Identical Containers
28 Coins

1. Fill each container with a different number of coins. First with one coin, second with two coins, and so on.
2. Mix them up so that you no longer know the order of the weights.

Activity	Time	Description
Launch	5 mins	The teacher poses the question to the students and invites a couple of students to try their ideas.
Problem-solving and algorithm	20 mins	Students try out and discuss different ways with each other. Student Worksheet: https://tinyurl.com/sortingactivityworksheet As the teacher progresses through the activities, the students progress along with the worksheet.
Discussions and Explorations	5 mins	The discussion on the various algorithms, how a slight change in approach leads to a different solution, and how computers apply such algorithms

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CT Components

This activity develops in a way that helps students discover important ideas, such as breaking a problem into smaller problems (decomposition), identifying patterns, developing algorithms, and finding faster ways to complete a task, thus analysing algorithms. These are fundamental ideas involved in computational thinking.

Activity: Sorting with Weighing Scale

Imagine we have 7 containers of different weights

We have to sort the containers by increasing weight using only a weighing scale (and nothing else). Our goal is to sort them using the fewest weighings.

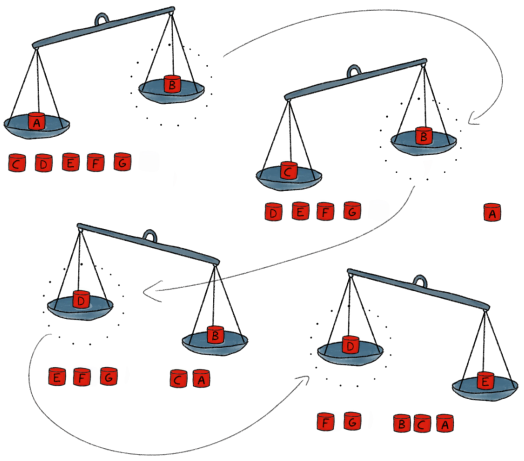




How will we figure this out? What process will we use to determine the minimum number of weighings needed to sort the containers by weight? Make sure that your way also works with the worst luck.

Method 1:

We compare 2 weights at a time to determine which is lighter. We compare it to the next container, then the next, and so on, until we find the lightest of the 7 containers. Then, from the remaining 6, find the lightest again, and so on.



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Q. By method 1, how many minimum weighings will it take to guarantee figuring out the lightest container of the 7? (This means we are assuming you always have bad luck and get the lightest one in the last possible step)

A. 4 weighings
B. 6 weighings
C. 7 weighings
D. 8 weighings

Answer: B. 6 weighings
Explanation: In the worst-case scenario, where we get the lightest weight, we compare it with the other 6 weights. Notice that even if we didn't start with the lightest, we still have to make 6 comparisons.
CT Component: Decomposition, algorithmic thinking,

Q. After finding the lightest of the 7 containers, you put it on the side. How many minimum weighings will it take to guarantee figuring out the lightest container of the remaining 6, (Always assume the worst luck).

A. 3 weighings
B. 6 weighings
C. 5 weighings
D. 7 weighings

Answer: C. 5 weighings
Explanation: Need to compare a total of five times. Reasoning is the same as the above problem.
CT Component: Decomposition, algorithmic thinking,

Q. If we continue this process of searching for the lightest in the remaining lot (by comparing 2 containers at a time), how many minimum weighings will it take to guarantee figuring out the sorting of the containers on the basis of their weight?

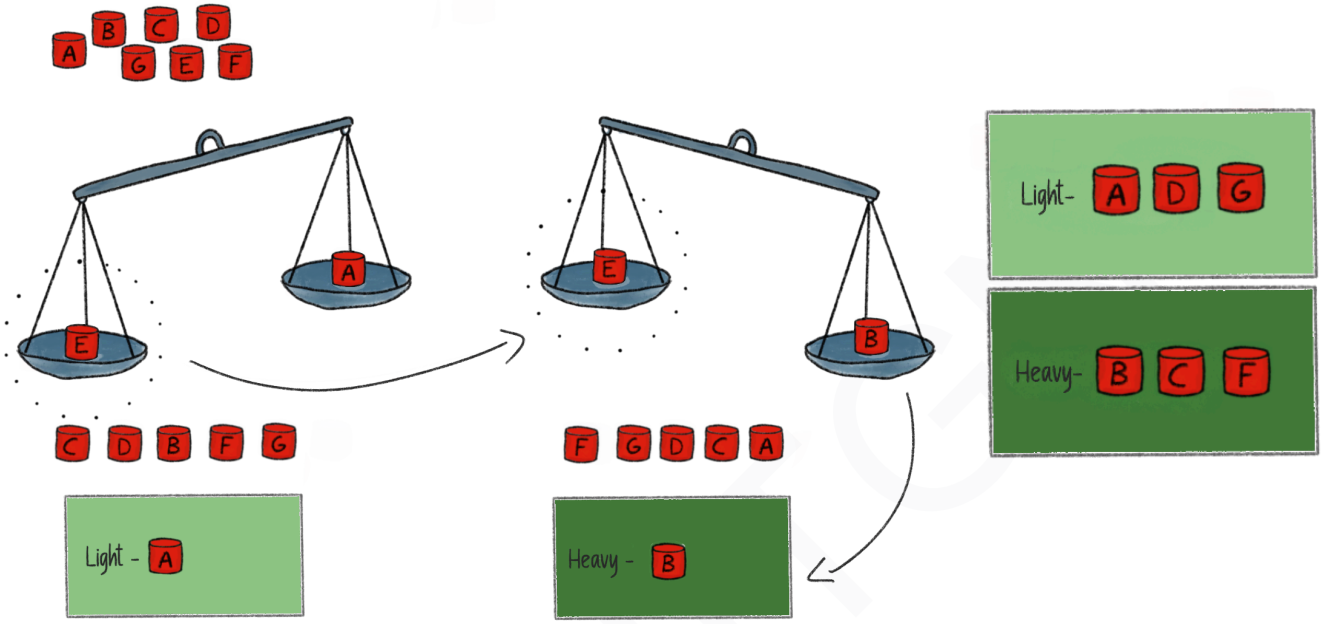
A. 42 weighings
B. 24 weighings
C. 28 weighings
D. 21 weighings

Answer: D.
Explanation: To figure out the lightest of the 7, it took 6 weighings; to figure out the lightest of the 6, it took 6 weighings; then 5, 4, 3, 2, and finally 1. If we add all this up, we get a total of 21 weighings.
CT Component: Pattern recognition, algorithmic thinking

Method 2: A way of random picking:
Another way could be to pick one container at random and compare all the others to it. Make 2 separate groups for the lighter and heavier containers, comparing them. And further do the same process for the groups formed.

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For example, we have 7 containers, randomly choose one, say E, and compare it to all the others one by one. First, we compare it with A; A is lighter than E; now compare E with B; B is heavier, and so on, comparing E with C, D, E, F, G one by one. A, D, G came lighter to E, whereas B, C, F came heavier to E.



Q. If we align all the 7 containers from lightest to heaviest _ _ _ _ _ _ _ (where 1st means lightest and 7th means heaviest), what will be the position of E?

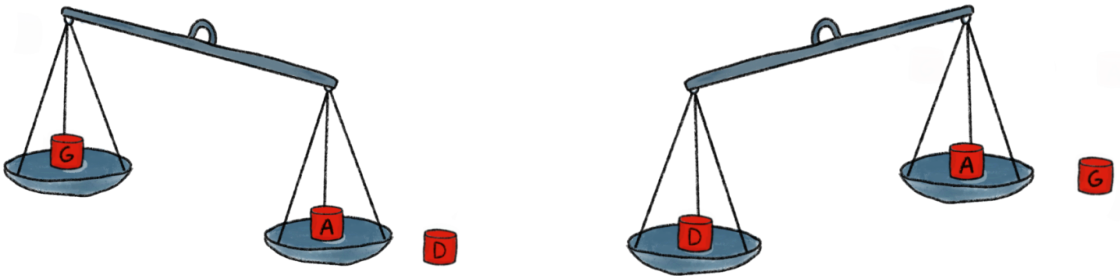
A. 4th
B. 3rd
C. 5th
D. Information is insufficient to say

Answer: A. 4th

Explanation: Three weights are lighter than E: A, D, G. Therefore, the position of E is 4 th from the beginning.

CT Component: Pattern recognition, Logical reasoning

Comparing all containers with E will fix the position for E. Now, for further sorting, we take all the light ones, which are A, D, and G. Choose one from the (say A) and compare it with the others, D and G.



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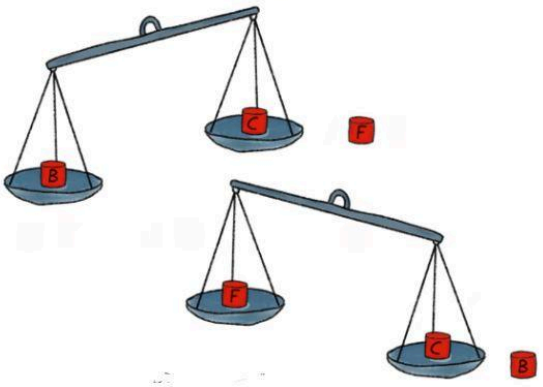
A is heavier than G, and A is lighter than D.

Q. What is the sorted order of A, G, D from lightest to heaviest?

A. D A G
B. G A D
C. A G D
D. A D G

Answer: B. G A D
Explanation: Weight of G < weight of A < weight of D
CT Component: Logical reasoning, algorithmic thinking

Similarly, for the group of containers heavier than E, which were B, C, and F, we chose C and compared it with the others to C and F.



C is lighter than B, and C is heavier than F.

Q. What is the sorted order of B, C, F from heaviest to lightest?

A. F C B
B. B F C
C. F B C
D. B C F

Answer: D. B C F
Explanation: weight of B > weight of C > weight of F based on the diagram

Q. What is the sorted order of A, B, C, D, E, F, G from lightest to heaviest?

A. G A D E F C B
B. A G D E F B C
C. F C B E G A D
D. Can not tell from the given information

Answer: C. F C B E G A D
Explanation: Putting together the results of the previous three questions.

Q. Total, how many weighings did it take to sort using method 2?

A. 4

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B. 5
C. 8
D. 10

Answer: D. 10

Explanation: To determine the position of weight E, 6 weighings were required. Then, to determine the positions of A and C, we took 2 weighings each, for a total of 10 weighings.

Q. How many weighings will it take to sort 5 weights in increasing order of their weights using method 1?

A. 10
B. 6
C. 15
D. 3

Answer: A. 10 weighings

Explanation: To find the lightest, it will take 4 weighings, then 3, then 2, then 1, for a total of 10 weighings.

CT Component: Pattern recognition, Algorithmic thinking, logical reasoning, decomposition

Q. If we do the same sorting of 5 weights using method 2, will the number of weighings be more than method 1?

A. Yes, it involves more steps
B. No, it will be less than method 1
C. Doesn’t make a difference

Answer: B. No, it will be less than method 1

Explanation: Method 2 breaks the problem into half-sized subproblems and keeps breaking it into smaller bits, each time by a factor of 1/2. In method one, each step decreases the number of steps by only one, so it takes much longer.

CT Component: Algorithmic thinking, decomposition, pattern recognition

Conclusion

Sorting objects can be done using different methods, each with its own efficiency. Some methods take more steps, while others reduce effort by organising comparisons smartly.

We learned that choosing the right approach depends on the task's goal.

Breaking the problem into smaller parts helps in solving it more effectively.

This activity shows that efficient thinking is important in both mathematics and computing.

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